QUICK GUIDE: SOLAR Meshcore LoRa REPEATER

By reading this guide, you will be able to acquire the necessary knowledge to carry out the check of the basic initial configuration, the first start-up, and the use of the Solar Meshcore LoRa Repeater.

Attention! You must first connect the included antenna while the device is turned off.

Warning: the product **contains flammable elements**, do not tamper with it, risk of explosion!

Topics covered in the guide:

1. Features list
2. General principle of operation
3. Initial assembly and start-up
4. Browsing of initial configuration
5. Start-up and use

Features list	
Heltec T114 v2.0 Meshcore LoRa board	■
Connectivity: LoRa, USBc	■
3,7v 4000 mA LiPo Battery with BMS	■
Battery thermal insulation	■
SX1262 LoRa Transceiver	■
Max LoRa transmit power +22 dBm	■
Hi Gain External antenna	■
Tiltable 6W Solar panel	■
IP67 case with pole bracket	■
GPS/GNSS module	optional
Board display	optional

General principle of operation

This device allows you to extend the radio coverage of any Meshcore LoRa terminal within its coverage range: it will act as an active repeater, bidirectionally forwarding packets from client devices and/or other connected repeaters.

Its internal battery is automatically recharged by the solar panel, ensuring service continuity even at night or on days with low solar radiation.
It should be installed in an elevated position (terraces, roofs, etc.) free from obstacles and will create an "umbrella" of radio frequency coverage with an indicative radius of up to a few kilometers (RPT to road user) or much greater if connected to the Mesh.

No electrical connection is required; the solar repeater is autonomous once permanently installed.
It also features a USB-C port for manually charging the battery.

Two or more LoRa-enabled terminals can communicate with each other without the aid of other technologies (WiFi, cellular networks, etc.) as long as they are located within the range of the repeater (a few kilometers) or using the "Meshed" network principle.

It goes without saying that through this "store & forward" mechanism, the Meshed network allows a theoretical extension of the range up to several dozen kilometers.

The services available in this area are:

• encrypted short messaging: the ability to exchange short messages between participants in a specific group or directly between two terminals; the messages are decrypted only by those terminals, as they hold the appropriate encryption key.

• real-time group location: the position of group members can be automatically exchanged and displayed on a map.

Initial assembly and start-up

Attention! You must first connect the included antenna while the device is turned off, then proceed pressing RST button to power it ON.

- Charge the internal battery to at least 30% using the USB-C service port on the bottom of the box if needed.

Tip: POWER OFF: in Meshcore FW release v1.18 + just connect to console and send “shutdown” command to power it OFF. Just unplug the battery alternatively.

- Close the cover by snapping the clips into place.
- Secure the solar panel bracket by rotating the ring nut on the support.
- Check the repeater configuration (see next paragraph) and perform initial connection tests with the client terminals (companion).
- Install the repeater in an open, elevated location free of obstacles.
- Orient the solar panel to the SOUTH at a 45-degree angle.





Management of initial configuration

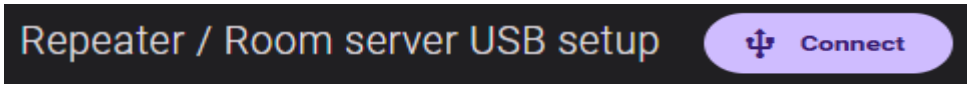
The Solar LoRa Mescore Repeater comes with a standard configuration based on the bandplan used (listed below).

Tip: Depending on the country/region of use, the LoRa Meshcore network configuration could have its own specifics, which must be known to access.

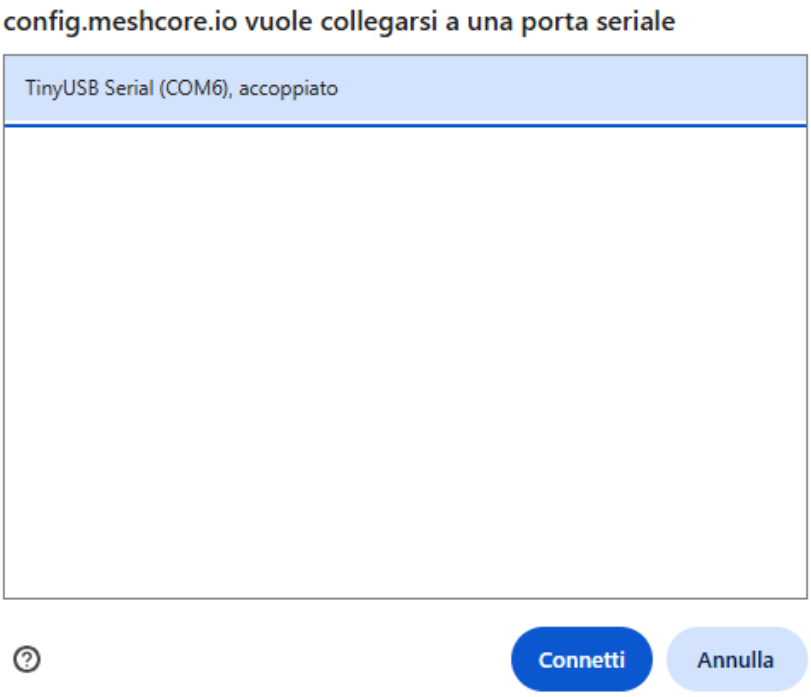
1) To configure the device, you must connect an online PC to the USBc port (the one directly attached to the Heltec board, not the one outside the Box, manage with care!)

2) Access the configuration menu, available at:

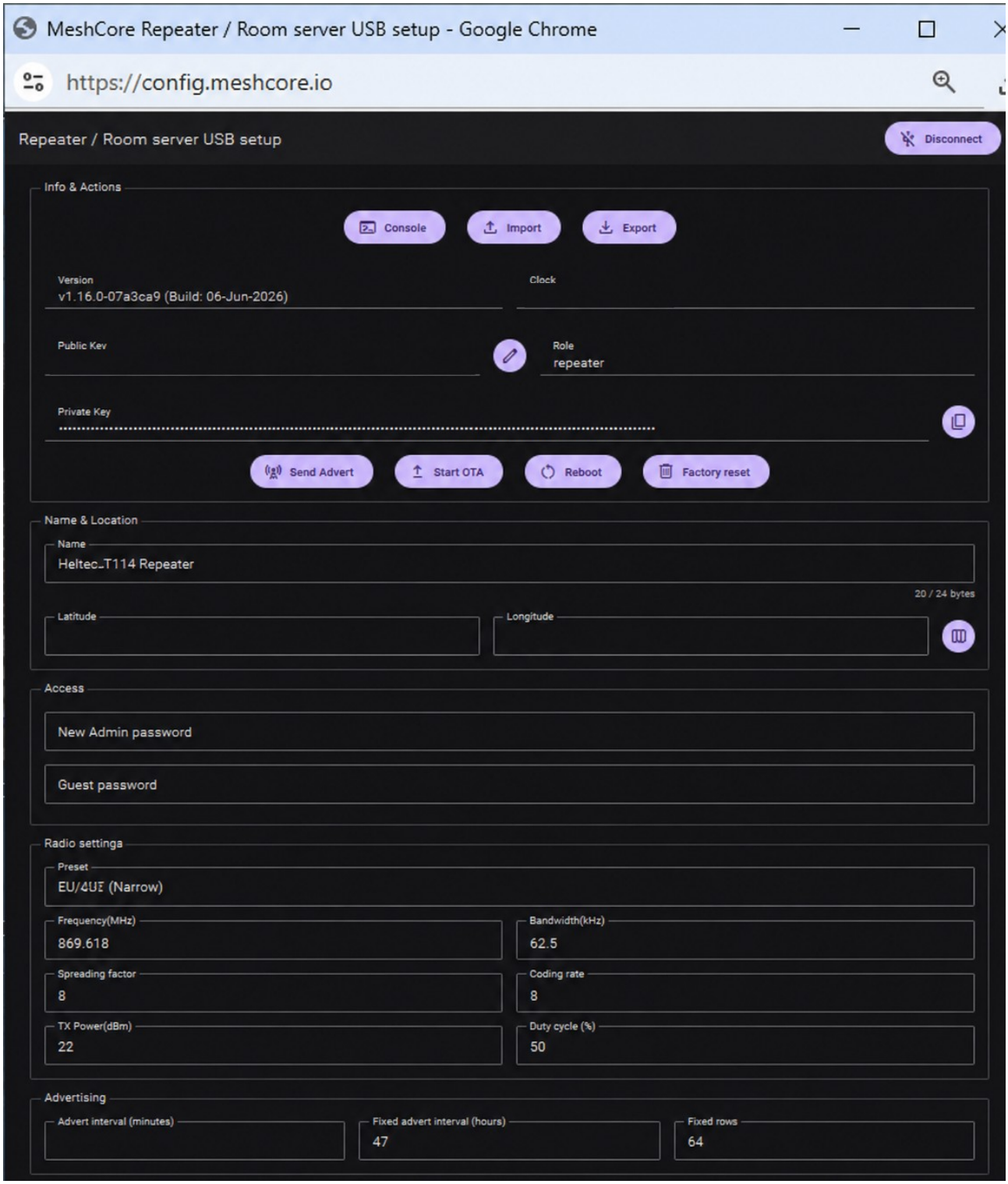
https://config.meshcore.io/



3) press Connect and select the shown USB interface then proceed:

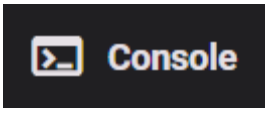


4) browse and check the running editable configuration:



Start up and use

While the Repeater is on and still connected to the PC via USBc, it is possible to invoke the System Console to check basic operation, as follows:

1) **https://meshcore.io/flasher**
2) **Tap Console button** 

3) **Will obtain the following screen:**



4) A bunch of low level commands is available, handle with care:

- **Ver** to check the installed FW
- **Clock** current time
- **Advert** to send an advert on the mesh: double check if some companion unit will discover it
- **Reboot** to reboot the repeater
- **Shutdown** to power OFF the repeater (FW rel v1.18+)